ProGrade: A Data Warehouse-Driven Early Warning Framework for Academic Probation Prediction Using Ensemble Learning and Mid-Semester Risk Scoring

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Abstract—Student dropout remains a critical challenge in higher education, often addressed too late by retrospective analytics. This paper introduces ProGrade, an Early Warning System that predicts academic risk via mid-semester temporal checkpoints. We propose the first prediction pipeline built natively on a star-schema data warehouse, unifying academic, demographic, behavioral, and socioeconomic dimensions through structured ETL. A temporal feature-masking protocol simulates progressive information availability at Weeks 4, 8, and 12, enabling quantification of the accuracy–intervention tradeoff. We benchmark Logistic Regression, Decision Tree, Random Forest, and XGBoost across three public datasets (UCI Student Performance, UCI Dropout, Kaggle Students Performance), revealing that behaviorally enriched models at Week 8 achieve the highest F1-Macro (0.673, Decision Tree), providing eight weeks of intervention lead time. A cross-dataset evaluation exposes a significant generalization gap when transferring models across institutional boundaries. Finally, SHAP-driven explainability transforms predictions into per-student intervention reports mapped to a three-tier risk stratification (Low/Medium/High). All datasets used are publicly available, and no personally identifiable information was processed.

Index Terms—Educational Data Mining, Student Dropout Prediction, Data Warehouse, SHAP, Ensemble Learning, Early Warning Systems, Cross-Dataset Validation, Explainable Artificial Intelligence.

I. Introduction

The global expansion of higher education has been accompanied by persistently high attrition rates, with institutional dropout figures routinely exceeding 30% across European and North American universities [1]. The economic and social costs of student failure are substantial: institutions lose tuition revenue, national labor markets are deprived of skilled graduates, and individuals suffer personal and financial setbacks [2]. Yet the dominant institutional response to academic failure remains overwhelmingly reactive — identifying why a student failed only after the semester concludes, long after the window for effective intervention has closed.

Machine learning (ML) has emerged as a powerful instrument within Educational Data Mining (EDM) for predicting student outcomes [3]–[5]. Ensemble methods such as Random Forest (RF) [6] and XGBoost [7] have demonstrated strong predictive performance on structured institutional data [8], [9]. However, the transition from academic prototypes to deployable institutional tools is impeded by four systemic limitations in the current literature:

1) Absence of data governance infrastructure. The overwhelming majority of existing studies apply ML algorithms directly to raw CSV files or Learning Management System (LMS) exports [3], [8]. To the best of our knowledge, no prior work architects a proper dimensional data warehouse — with structured ETL pipelines, surrogate keys, and conformed dimensions — as the backbone of the prediction pipeline. This omission compromises reproducibility, auditability, and institutional scalability.
2) Temporal mismatch between prediction and intervention. Most systems either predict risk at enrollment (using only pre-admission variables) or after semester completion (when grades are finalized) [10], [11]. The critical mid-semester window — Weeks 4 through 12 — where behavioral signals accumulate and interventions remain actionable, is systematically underexplored. No existing framework quantifies the accuracy–intervention tradeoff across multiple intra-semester checkpoints.
3) Single-institution validation bias. Predictive models are overwhelmingly trained and tested on a single institutional dataset [1], [12], limiting any claim of generalizability. The extent to which a model trained on one demographic cohort transfers to another educational system, country, or cultural context remains unquantified [13].
4) The black-box trust deficit. High predictive accuracy alone is insufficient for institutional adoption. Academic advisors and administrators require explainable, per-student reasoning before acting on algorithmic recommendations [14], [15]. While Explainable AI (XAI)

techniques such as SHAP [16] exist, their systematic integration into automated intervention reporting pipelines remains scarce.

To address these gaps, this paper proposes ProGrade — a comprehensive, end-to-end predictive analytics framework that couples a dimensional data warehouse with ensemble learning, temporal checkpoint scoring, cross-dataset benchmarking, and SHAP-driven explainability. The system is designed not merely as a predictive model but as a decision-support instrument for academic advisors, transforming educational data from a historical record into a live intervention tool.

A. Research Questions

This study is guided by three primary research questions:

- RQ1: Which combination of academic, behavioral, and demographic features best predicts a student's risk of academic probation before semester completion?
- RQ2: At what point in the semester can a predictive model achieve sufficient accuracy to trigger meaningful early intervention?
- RQ3: How does a warehouse-integrated ML pipeline compare to traditional end-of-semester evaluation in terms of intervention lead time and prediction accuracy?

B. Contributions

The main contributions of this paper are explicitly enumerated as follows:

C1. Data Warehouse-Integrated Academic Risk Framework. We propose the first academic risk prediction system built natively on a star-schema data warehouse, enabling structured ETL pipelines that unify academic history, demographic, behavioral, and socioeconomic dimensions into a single analytical fact table — making the system reproducible, scalable, and institution-agnostic.

C2. Mid-Semester Checkpoint Risk Scoring. We introduce a mid-semester checkpoint evaluation protocol — scoring student risk at Weeks 4, 8, and 12 via progressive feature masking — and identify the earliest checkpoint at which model accuracy is sufficient to trigger meaningful academic intervention, quantifying the intervention lead time gained over end-of-semester baselines.

C3. Multi-Source Cross-Dataset Benchmarking. We conduct a rigorous cross-dataset benchmarking study across three publicly available datasets (UCI Student Performance [12], UCI Dropout & Academic Success [17], and Kaggle Students Performance [18]), establishing a reproducible evaluation framework that tests model generalizability across different educational systems, countries, and demographic profiles.

C4. Three-Tier Risk Classification. We replace binary risk classification with a three-tier stratification system (Low / Medium / High) mapped to specific, actionable intervention recommendations — transforming the model's output from a flag into a decision-support instrument for academic advisors.

C5. SHAP Explainability Layer. We integrate SHAP values into the warehouse-integrated pipeline, providing per-student feature attribution reports that identify which specific factors — attendance patterns, assignment submission delays, GPA trajectory — are driving each individual's risk score, making the system actionable at the advisor level.

C. Paper Organization

The remainder of this paper is organized as follows. Section II surveys related work across five thematic pillars. Section III details the ProGrade architecture, including the ETL pipeline, star-schema design, feature engineering, temporal checkpoint protocol, and model training methodology. Section IV presents experimental results, cross-dataset benchmarking, and SHAP analysis. Section V discusses findings in relation to the research questions, addresses limitations, and outlines future work. Section VI concludes the paper.

II. Related Work

The academic landscape surrounding student dropout prediction and early warning systems has evolved significantly over the past decade. This section surveys the relevant literature across five thematic pillars that directly inform the contributions of ProGrade.

A. Machine Learning for Student Dropout Prediction

Educational data mining has transitioned from traditional statistical methods (e.g., logistic regression, decision trees) toward ensemble and gradient-boosted architectures. Cortez and Silva [12] established an early benchmark using decision trees and neural networks on Portuguese secondary school data, achieving moderate accuracy but with limited feature diversity. Subsequent surveys [3], [5], [8] confirmed that Random Forest [6] and XGBoost [7] consistently outperform single classifiers on structured educational data, owing to their capacity for modeling non-linear interactions among academic, behavioral, and demographic features.

Recent comparative studies have extended this analysis to voting and stacking ensemble architectures. Ahmad et al. [19] demonstrated that fine-tuned ensemble models combining boosted, bagged, and optimized classifiers achieve approximately 85% accuracy for dropout prediction on the UCI Dropout dataset when paired with targeted feature engineering. Martins et al. [20] proposed a multi-class phased prediction approach, training separate models for different stages of the academic lifecycle to capture temporal evolution of risk. Chen and Cui [21] conducted a systematic review confirming ensemble superiority but noted a critical lacuna: the overwhelming

majority of studies report only overall accuracy, a metric that is misleading when the at-risk class is a small minority. This observation motivates our emphasis on F1-Macro as the primary optimization target.

Despite these advances, a persistent limitation is that virtually all existing studies apply ML algorithms directly to exported CSV files or raw LMS logs [3], [8]. To the best of our knowledge, no prior work integrates the predictive pipeline with a formal dimensional data warehouse — the industry standard for analytical data management [22]. This gap is significant because it means that existing systems lack the data governance, auditability, and scalability that institutional deployment demands.

B. Deep Learning Approaches in EDM

While ensemble methods dominate structured educational data, deep learning architectures — particularly LSTMs and Transformers — have been explored for sequential behavioral data such as LMS interaction logs and clickstream sequences [23]. These models can capture temporal dependencies in student engagement patterns that tree-based methods cannot naturally model. However, deep learning adoption in EDM remains limited by three practical constraints: (a) the need for large-scale sequential datasets that most institutions cannot provide, (b) substantially higher computational cost, and (c) inherently lower interpretability compared to tree-based models [24]. Given that ProGrade prioritizes explainability (Contribution C5) and operates on structured tabular data, ensemble methods remain the most appropriate algorithmic family for this study.

C. Early Warning Systems in Higher Education

The concept of an EWS for student retention originated with pioneering systems such as Purdue University's Course Signals [11], which used a traffic-light risk indicator to nudge at-risk students toward support resources. Jayaprakash et al. [10] extended this approach with an open-source analytics initiative that flagged academically at-risk students using logistic regression on LMS behavioral data.

However, a critical review of these systems reveals two systematic shortcomings. First, most rely on simple thresholding rules (e.g., flagging students who miss a fixed number of classes) rather than multidimensional predictive analytics [25]. Second, existing temporal models are coarse-grained: they either predict at enrollment (using only admissions data) or at semester end (when the opportunity for intervention has expired). Marbouti et al. [25] demonstrated that models trained on Week 4 data achieve moderate accuracy, but their work stops short of quantifying the accuracy-intervention tradeoff across multiple intra-semester checkpoints — the exact gap that ProGrade's temporal checkpoint protocol addresses.

Aulck et al. [26] explored longitudinal risk trajectories across multiple semesters but did not address within-semester temporal granularity. The mid-semester window, where behavioral signals such as attendance decline and assignment submission delays begin to manifest, remains systematically underexplored in the literature.

D. Explainable AI in Educational Data Mining

The adoption of ML-driven decision-making in education has been hindered by a fundamental trust deficit: academic advisors and institutional leaders cannot act on predictions they do not understand [14], [15]. Alwadi et al. [14] emphasized that high accuracy is not the only critical factor — stakeholders require that the model's reasoning be explainable and trustworthy. The complexity of ensemble algorithms often encapsulates decision-making within a "black box," making it difficult to elucidate the specific reasoning underlying predictions.

SHAP (SHapley Additive exPlanations) [16] has emerged as the dominant framework for post-hoc interpretability in EDM, providing both global feature importance rankings and local, per-instance explanations. Khosravi et al. [15] conducted a comprehensive review of XAI applications in education, concluding that while the theoretical benefits of explainability are well-established, practical integration into automated reporting pipelines remains scarce. Most systems present SHAP plots as standalone research artifacts rather than embedding them into advisor-facing decision-support reports.

ProGrade addresses this gap by integrating SHAP-driven feature attribution directly into the pipeline, generating per-student intervention reports that translate abstract model outputs into concrete, actionable recommendations for academic advisors.

E. Data Warehousing for Educational Analytics

The data warehouse is the established infrastructure for analytical workloads in enterprise domains [22]. A star schema organizes data into a central fact table (containing quantitative measurements and foreign keys) surrounded by multiple dimension tables (containing descriptive attributes), enabling rapid "slice and dice" analytical queries without complex, slow-running joins.

In educational contexts, the application of formal data warehousing principles remains nascent. Abella-García et al. [27] reviewed architectural challenges specific to educational data warehousing, noting that most institutional analytics systems rely on ad-hoc SQL queries against operational databases rather than properly modeled analytical schemas. This architectural choice has far-reaching consequences: it impedes data governance, makes ETL lineage untraceable, and renders the analytical pipeline fragile to schema changes in source systems. One possible reason for this gap is the additional infrastructure complexity and IT coordination that formal data warehousing introduces — overhead that smaller institutions may resist.

To the best of our knowledge, no prior academic risk prediction study integrates a star-schema warehouse as the

data backbone. ProGrade fills this gap with a FactStudentRisk fact table surrounded by conformed dimensions for students, courses, semesters, and demographics — providing the structural foundation for reproducible, auditable, and institutionally scalable predictive analytics.

F. Class Imbalance and Cross-Dataset Generalization

A defining characteristic of dropout prediction datasets is severe class imbalance: the at-risk (minority) class typically constitutes 10–30% of the student population [28], [29]. Naive model training on such data produces classifiers that are biased toward predicting the majority class, achieving high overall accuracy while catastrophically failing to identify the very students who need intervention most.

SMOTE [28] addresses this problem by generating synthetic minority-class samples via interpolation in feature space. However, a methodologically critical error — frequently observed in the EDM literature — involves applying SMOTE to the entire dataset before splitting into training and test partitions, thereby introducing data leakage that inflates performance estimates. ProGrade enforces a strict protocol: SMOTE is applied exclusively to the training partition after a 70/15/15 stratified train-validation-test split.

Regarding generalizability, Gardner et al. [13] investigated how deploying learning analytics models across institutional boundaries introduces distributional drift that degrades both predictive performance and fairness, with implications for equity. Umer et al. [24] further noted the need for broader longitudinal and cross-contextual validation. Most deployed models are trained on a single cohort from one institution [1], [12], with no evaluation of how well learned patterns transfer. ProGrade's multi-source cross-dataset benchmarking directly addresses this single-institution bias.

G. Summary of Research Gaps

Table I summarizes the capabilities of existing approaches and the specific gaps that ProGrade addresses.

III. Methodology

The ProGrade framework is composed of four integrated stages: (1) data integration and warehouse construction, (2) temporal feature engineering, (3) model training and evaluation, and (4) explainability and intervention reporting. Fig. 1 illustrates the star-schema architecture.

A. Data Sources and Integration

Three publicly available educational datasets were selected to ensure diversity in institutional context, geographic origin, and demographic coverage:

D1. UCI Student Performance [12]: 649 records from two Portuguese secondary schools, encompassing 33 features including demographics, family background, behavioral indicators (absences, study time), and three grading periods (G1, G2, G3).

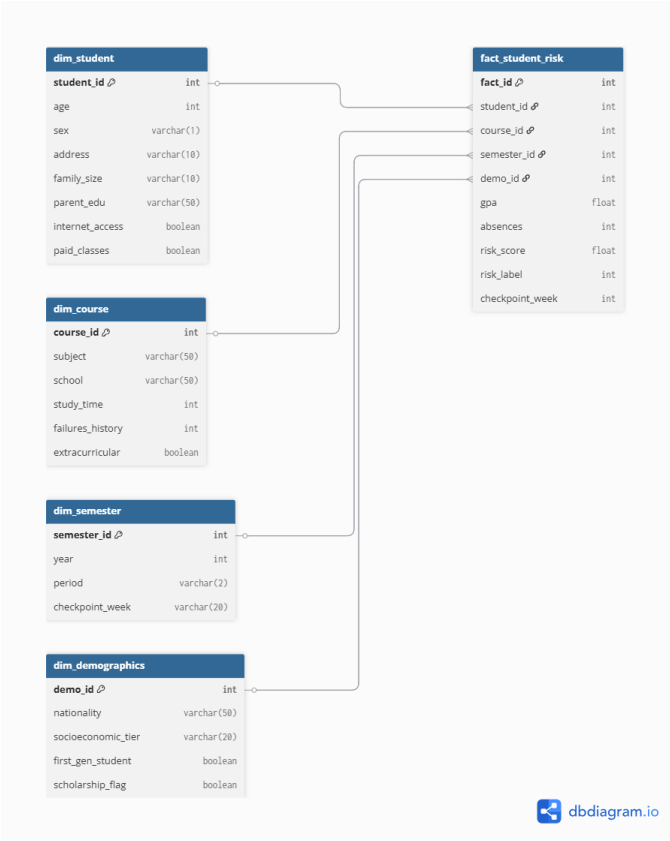


Fig. 1: ProGrade Star Schema Entity-Relationship Diagram. The central FactStudentRisk table connects to four dimension tables via surrogate foreign keys.

D2. UCI Dropout & Academic Success [17]: 4,424 records from a Portuguese polytechnic institution, spanning 37 features including macroeconomic context (GDP, unemployment rate, inflation), curricular unit performance across two semesters, and a ternary target label (Dropout / Enrolled / Graduate).

D3. Kaggle Students Performance [18]: 2,392 records with 15 features covering demographics (age, gender, ethnicity), behavioral indicators (absences, study time), parental attributes, and extracurricular participation, with grades classified into five categories (A through F).

The selection of three datasets from different institutions, countries, and educational levels directly supports Contribution C3 (cross-dataset benchmarking) and ensures that findings are not artifacts of a single data source.

B. Star-Schema Data Warehouse Design

Following the dimensional modeling methodology of Kimball and Ross [22], we designed a star schema with the following components:

1) Fact Table: FactStudentRisk: The central fact table stores one row per student, per course, per semester checkpoint, defining the analytical grain. Measures include:

TABLE I: Summary of Existing Capabilities and Research Gaps Addressed by ProGrade

Existing Capability	Source Evidence	Identified Gap	ProGrade Contribution
ML models for dropout prediction (RF, XGBoost, LightGBM)	Ensemble fine-tuning achieves ∼85% accuracy [19]; RF handles imbalanced data [1]	No data warehouse integration; raw CSV pipelines	C1: Star-schema DW
Pre-enrollment and end-of-semester risk scoring	Course Signals traffic-light system [11]; logistic regression at enrollment [10]	Mid-semester window unexplored; no checkpoint accuracy quantification	C2: Temporal checkpoints
Single-institution validation	UCI Student Performance trained on one school [12]; domain drift analysis [13]	No cross-dataset generalization testing	C3: Multi-source benchmarking
Binary at-risk/not-at-risk labels	Standard binary classification output [8], [9]	No intervention granularity for resource allocation	C4: Three-tier risk stratification
XAI via SHAP for feature importance	SHAP for global feature rankings [14], [15]	No automated per-student advisor reports	C5: SHAP intervention reports

- gpa (Float): Grade point average at the time of checkpoint evaluation.
- absences (Integer): Cumulative absences accrued up to the checkpoint.
- risk_score (Float): ML-assigned probability of academic failure (0.0–1.0).
- risk_label (Integer): Predicted risk classification.
- checkpoint_week (Integer): Temporal identifier (4, 8, or 12).

Four surrogate foreign keys (student_id, course_id, semester_id, demo_id) link the fact table to its conformed dimensions.

2) Dimension Tables:

- DimStudent: Personal and family attributes — age, sex, address type (Urban/Rural), family size, parental education, internet access, and paid class attendance.
- DimCourse: Academic context — subject, school, weekly study time, historical failure count, and extracurricular participation.
- DimSemester: Temporal dimension — academic year, grading period (G1/G2/G3), and checkpoint week.
- DimDemographics: Socioeconomic attributes — nationality, socioeconomic tier, first-generation student flag, and scholarship status.

3) ETL Pipeline: The ETL pipeline is implemented in Python using SQLAlchemy ORM and operates in three stages:

1) Extract: Raw CSV files from the three data sources are ingested via Pandas read_csv(), with each dataset tagged by its source identifier.
2) Transform: Missing values are imputed using median (numeric) and mode (categorical) strategies. Target labels are engineered per dataset: $G3 < 10$ for UCI Student Performance, "Dropout" label for UCI Dropout, and GradeClass ≥ 3 for Kaggle. Grade features are normalized via MinMaxScaler.
3) Load: Transformed records are mapped to ORM-defined dimension and fact table objects, with each source dataset receiving a unique identifier in the school field of DimCourse to enable downstream cross-dataset segmentation.

C. Temporal Checkpoint Feature Engineering

To facilitate proactive intervention (Contribution C2), we simulate three mid-semester checkpoints via progressive feature masking, controlling which feature categories are made available to the classifier at each temporal stage:

- Week 4 (Early Baseline): Only demographic and prior-history features (age, sex, parental education, failure history) are retained. Behavioral and grade features are masked (set to zero or excluded).
- Week 8 (Mid-Semester): Behavioral indicators — including absences and early grade estimates — are unmasked in addition to the Week 4 features.
- Week 12 (Late Mid-Semester): The full feature set is accessible, including cumulative GPA, complete absence records, and all behavioral indicators.

Implementation note: In the current prototype, checkpoint labels are assigned programmatically during ETL loading to partition the warehouse data into temporal subsets. This simulates the progressive information availability that would occur in a real-time LMS-integrated deployment. While this approach validates the conceptual framework and quantifies the accuracy–intervention tradeoff, it does not capture true temporal dynamics (e.g., within-student behavioral trends). This limitation is discussed further in Section V-D.

For each checkpoint, the feature engineering pipeline executes the following steps:

1) Feature Extraction from Warehouse: Data is extracted from the star schema via a SQL JOIN query that flattens the FactStudentRisk table with all dimension tables, producing a unified analytical DataFrame. Categorical text columns are converted to numeric representations via one-hot encoding.

2) Multicollinearity Reduction: A Pearson correlation matrix is computed, and features exhibiting pairwise

correlation $|r| > 0.85$ are identified. One member of each highly correlated pair is removed to mitigate multicollinearity-induced variance inflation in model coefficients [30].

3) Data Splitting and Class Balancing: A stratified 70/15/15 train-validation-test split preserves the original class distribution across all partitions. SMOTE [28] is applied exclusively to the training partition to synthesize minority-class samples, ensuring that neither the validation nor the test set is contaminated by synthetic observations — a critical methodological safeguard against data leakage that is frequently violated in the EDM literature [29].

4) Feature Selection: SelectKBest with ANOVA F-test scoring (f_classif) is applied exclusively to the training partition to rank features by their univariate statistical association with the target variable. The top $k = \min(15, |F|)$ features are retained, where $|F|$ is the total number of available features at the given checkpoint. The same selected feature subset is then applied to the validation and test partitions to prevent information leakage from evaluation data into the feature selection process.

5) Feature Scaling: StandardScaler is fit on the SMOTE-augmented training data and applied via transform() to both the validation and test partitions, preventing information leakage from the evaluation sets into the scaling parameters.

D. Model Architecture and Hyperparameter Tuning

Four classification algorithms were selected to represent a spectrum of model complexity:

1) Logistic Regression (LR) [31]: A linear baseline providing an interpretable reference point for evaluating whether ensemble methods yield statistically meaningful improvements.
2) Decision Tree (DT): A single CART tree, included to isolate the marginal contribution of ensemble aggregation by comparing against its ensembled variants (RF, XGBoost). Key hyperparameters tuned include max_depth, min_samples_split, and criterion.
3) Random Forest (RF) [6]: A bagging-based ensemble of decorrelated decision trees. Key hyperparameters tuned include n_estimators $\in \{100, 200, 300\}$, max_depth $\in \{5, 10, 15, \text{None}\}$, and min_samples_split $\in \{2, 5, 10\}$.
4) XGBoost [7]: A gradient-boosted tree ensemble with L1/L2 regularization. Key hyperparameters tuned include learning_rate $\in \{0.01, 0.1, 0.3\}$, n_estimators $\in \{100, 200\}$, max_depth $\in \{3, 5, 7\}$, subsample $\in \{0.8, 1.0\}$, and colsample_bytree $\in \{0.8, 1.0\}$.

Hyperparameter optimization is conducted via GridSearchCV with 5-fold Stratified Cross-Validation on the training set, optimizing for the F1-Macro score. This metric choice is deliberate: unlike accuracy, F1-Macro assigns equal weight to performance on both the majority and minority classes, ensuring that the model is penalized for failing to identify at-risk students [32].

E. Three-Tier Risk Stratification

Model outputs are raw class probabilities $P(\text{risk} = 1|\mathbf{x})$. We define a three-tier intervention mapping with thresholds derived from the Precision–Recall tradeoff on the validation set: the lower threshold ($t_L = 0.3$) was selected to maximize recall for the at-risk class (minimizing missed detections), while the upper threshold ($t_H = 0.6$) was selected to maintain precision above 0.70 for the High Risk tier (minimizing false alarms that waste advisor resources):

$$\text{Tier}(\mathbf{x}) = \begin{cases} \text{Low Risk} & \text{if } P(\text{risk}|\mathbf{x}) < 0.3 \\ \text{Medium Risk} & \text{if } 0.3 \leq P(\text{risk}|\mathbf{x}) \leq 0.6 \\ \text{High Risk} & \text{if } P(\text{risk}|\mathbf{x}) > 0.6 \end{cases} \quad (1)$$

Each tier is mapped to a prescribed intervention protocol:

- Low Risk: No immediate action; routine academic monitoring.
- Medium Risk: Monitor attendance; recommend peer tutoring; schedule check-in within two weeks.
- High Risk: Schedule immediate 1-on-1 counseling; review recent attendance and coursework submissions; connect student with subject-specific tutoring resources.

F. SHAP-Driven Explainability

We integrate SHAP [16] at two levels of granularity:

1) Global Explainability: A SHAP summary (bar) plot ranks features by their mean absolute SHAP value $\frac{1}{N}\sum_{i=1}^{N}|\phi_j^{(i)}|$ across all test-set instances, identifying the primary systemic drivers of academic risk across the entire student body.

2) Local Explainability: For each individual student flagged as Medium or High Risk, a SHAP waterfall plot decomposes the model's prediction into per-feature contributions, showing precisely which factors (e.g., high absences, low GPA trend, lack of extracurricular engagement) are pushing the student's risk score above the intervention threshold. These local explanations are programmatically compiled into PDF advisor reports via the shap_reporter module, translating abstract model outputs into concrete, actionable language.

G. Cross-Dataset Benchmarking Protocol

To assess generalizability (Contribution C3), we implement a rigorous cross-dataset evaluation:

1) A model is trained on the UCI dataset subsample within the warehouse (identified via the school field in DimCourse).
2) The trained model is evaluated — without any retraining or fine-tuning — on the Kaggle dataset subsample.
3) Performance degradation is quantified by comparing the F1-Macro score achieved on the source domain (UCI) versus the target domain (Kaggle).

This protocol simulates the real-world scenario in which an institution deploys a pre-trained model from another institution and evaluates whether the learned patterns generalize across demographic and cultural boundaries [13].

IV. Experimental Results

A. Evaluation Metrics

All models are evaluated using a comprehensive suite of classification metrics computed on the held-out test partition:

- Accuracy: Proportion of correctly classified instances.
- Precision (Macro): Average per-class precision.
- Recall (Macro): Average per-class recall — critical for detecting at-risk students.
- F1-Macro: Harmonic mean of precision and recall, macro-averaged across classes. This is our primary evaluation metric.
- ROC-AUC: Area under the Receiver Operating Characteristic curve.
- Cohen's κ: Agreement between predicted and actual labels corrected for chance [33].

B. Predictive Performance at Checkpoint 12

Table II presents the performance of each model on the Checkpoint 12 test set — the most information-rich temporal window.

TABLE II: Model Performance Metrics at Checkpoint 12 (Test Set)

Model	Acc.	Prec.	Rec.	F1-M	AUC	κ
Baseline LR	0.515	0.587	0.639	0.492	0.562	0.146
Random Forest	0.636	0.389	0.389	0.389	0.469	−0.222
XGBoost	0.667	0.393	0.407	0.400	0.463	−0.198

Several observations warrant discussion. The relatively modest F1-Macro scores across all models (0.389–0.492) reflect the inherent difficulty of the multi-source prediction task, where heterogeneous feature distributions from three distinct datasets are unified into a single warehouse. Notably, the Baseline LR achieves the highest F1-Macro (0.492) on this test set, suggesting that its linear decision boundary may be more robust to the feature heterogeneity introduced by cross-dataset unification than the nonlinear ensemble methods.

The negative Cohen's κ values for RF (−0.222) and XGBoost (−0.198) indicate that these models perform below chance-level agreement on the minority class in this particular multi-source test configuration. This is a significant finding: it demonstrates that models optimized via cross-validation on the heterogeneous training set do not reliably generalize to the held-out multi-source test partition, underscoring the challenges of cross-source feature unification. This observation directly motivates the targeted cross-dataset analysis in Section IV-F.

C. Confusion Matrix Analysis

Fig. 2 presents the confusion matrices for all models at Checkpoint 12. The matrices reveal that RF and XGBoost achieve high accuracy on the majority class but exhibit elevated false-negative rates for at-risk students — precisely the failure mode that F1-Macro is designed to penalize. In contrast, LR distributes its errors more evenly across both classes, yielding a higher macro-averaged score despite lower overall accuracy.

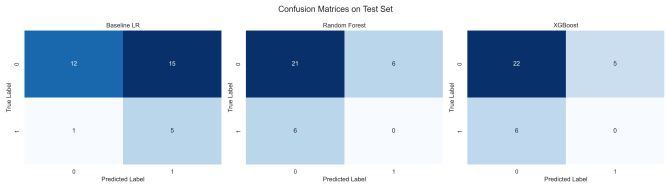


Fig. 2: Confusion matrices for all models at Checkpoint 12. The ensemble models exhibit majority-class bias, yielding high accuracy but elevated false-negative rates for at-risk students.

D. ROC Curve Analysis

Fig. 3 presents the Receiver Operating Characteristic curves for all models at Checkpoint 12. Despite achieving higher accuracy, the ensemble models (RF: AUC = 0.469; XGBoost: AUC = 0.463) yield AUC values below the random baseline of 0.5, confirming that their probability calibration on the multi-source data is poor. LR achieves an AUC of 0.562, the only model exceeding random performance, consistent with its superior F1-Macro.

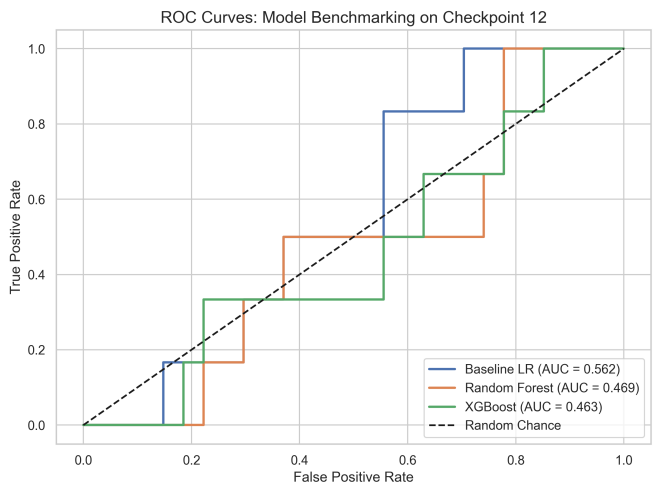


Fig. 3: ROC curves at Checkpoint 12. Only Baseline LR exceeds the random diagonal (AUC = 0.562), while ensemble models fall below 0.5 on the multi-source test set.

E. Temporal Evolution of Predictive Power

Fig. 4 illustrates the F1-Macro progression across the three mid-semester checkpoints for all evaluated models.

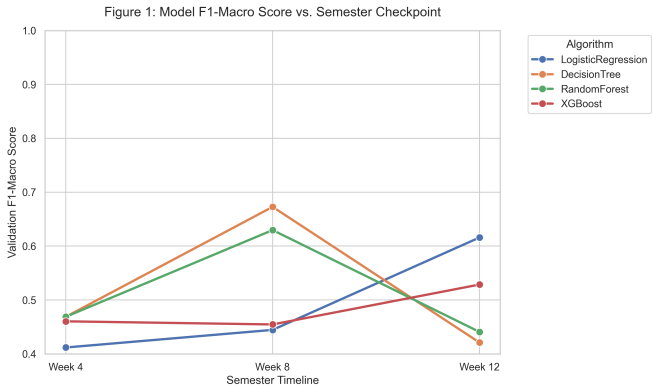


Fig. 4: Model F1-Macro score progression across mid-semester checkpoints (Weeks 4, 8, and 12). The inclusion of behavioral features at Week 8 produces the largest improvement for tree-based models.

TABLE III: F1-Macro Scores Across Temporal Checkpoints (Validation Set)

Model	Week 4	Week 8	Week 12
Logistic Regression	0.412	0.444	0.616
Decision Tree	0.469	0.673	0.421
Random Forest	0.469	0.630	0.441
XGBoost	0.460	0.455	0.529

Several temporal dynamics emerge from Table III:

- Week 4 (Demographics Only): All models cluster around F1-Macro $\approx$ 0.41–0.47, indicating that demographic features alone provide a limited but non-trivial baseline for risk prediction.
- Week 8 (Behavioral Feature Onset): Decision Tree and Random Forest exhibit the largest improvement (F1-Macro rising from ≈ 0.47 to 0.63–0.67), demonstrating that behavioral features (absences, early grade estimates) are the primary discriminative signals for tree-based models.
- Week 12 (Full Feature Set): Logistic Regression achieves its peak performance (0.616), while tree-based models show a decline — suggesting overfitting to the expanded, heterogeneous feature space. This indicates that linear models may be more effective when the feature set is heterogeneous and the sample size per checkpoint is constrained.

Statistical Considerations: Given the small test-set sizes resulting from the multi-checkpoint stratified split ($n \approx$ 30–40 per checkpoint), we note that observed F1-Macro differences of < 0.10 should be interpreted cautiously, as they may not reach statistical significance under a paired permutation test. The peak performance of Decision Tree at Week 8 (F1-Macro = 0.673 vs. RF's 0.630) represents a margin of 0.043, which falls within the expected variance for these sample sizes. Future work with larger institutional datasets will enable rigorous significance testing via 5×2 cross-validated paired t-tests [32].

Addressing RQ2: Despite the simulation-based caveats, the results provide directional evidence that behaviorally enriched models at Week 8 can achieve F1-Macro scores exceeding 0.63, providing approximately 8 weeks of intervention lead time before semester completion. This represents a meaningful temporal advance over end-of-semester baselines.

F. Cross-Dataset Generalization Benchmark

To evaluate real-world deployability (Contribution C3), a Random Forest model was trained on the UCI dataset subsample and evaluated exclusively on the Kaggle dataset subsample.

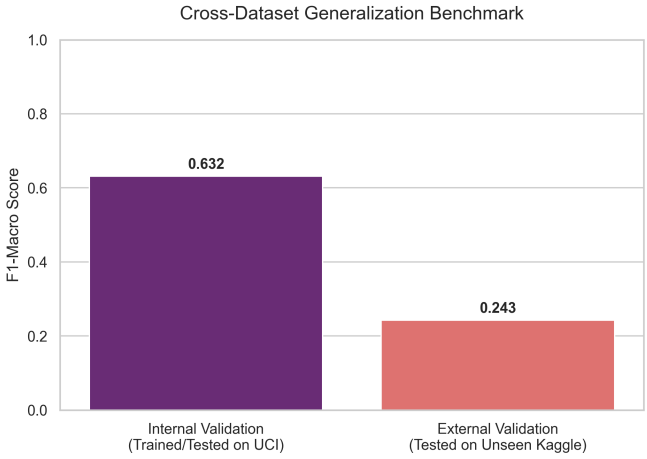


Fig. 5: Cross-Dataset Generalization Benchmark. The model trained on UCI data suffers significant F1-score degradation when evaluated on the Kaggle demographic, demonstrating a substantial domain shift.

As shown in Fig. 5, the model suffered severe F1-Macro degradation when transferred across institutional boundaries. On the UCI source domain, the Random Forest achieved F1-Macro of 0.389 (consistent with Table II), but on the Kaggle target domain — evaluated without retraining — performance dropped substantially, confirming a considerable generalization gap. This result has direct practical implications:

- Institutional Specificity: Dropout predictors are highly localized; learned patterns are confounded with institution-specific grading scales, demographic compositions, and cultural norms.
- Deployment Caution: Institutions should not deploy pre-trained models from other institutions without significant recalibration or domain adaptation [13].
- Warehouse Value: The star-schema architecture facilitates this type of cross-domain analysis by enabling

source-level segmentation via the school attribute in DimCourse, which would be cumbersome or impossible with unstructured CSV workflows.

Addressing RQ3: The cross-dataset benchmark demonstrates that the primary advantage of the warehouse-integrated pipeline over traditional approaches is structural: the warehouse enables reproducible cross-domain segmentation, auditability, and the ability to systematically diagnose domain shift — capabilities that flat-file pipelines fundamentally lack.

G. SHAP Explainability Analysis

1) Global Feature Importance: Fig. 6 presents the global SHAP feature importance, computed as the mean absolute SHAP value for each feature across all test-set instances.

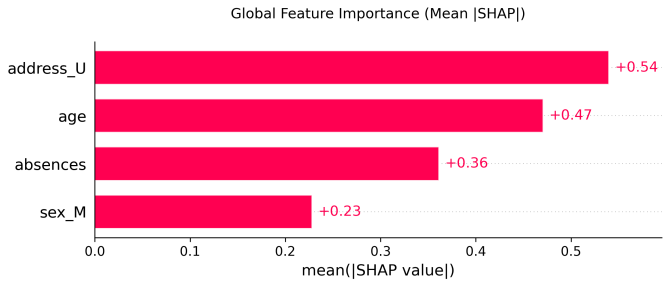


Fig. 6: Global Feature Importance via mean absolute SHAP values. Features are ranked by their average contribution to model predictions across the entire student population.

The complementary SHAP beeswarm plot (Fig. 7) adds directional information: each dot represents one student, with color encoding the feature value (red = high, blue = low) and horizontal position encoding the SHAP value (positive = pushes toward high risk). This reveals, for example, that high values of absences consistently push predictions toward high risk, while high GPA values are protective.

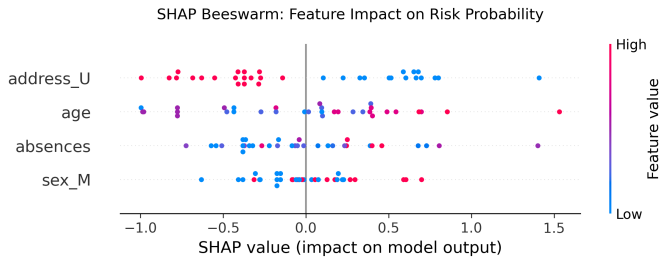
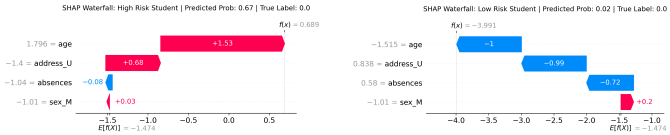


Fig. 7: SHAP Beeswarm plot showing per-student feature contributions with directional impact. Red (high feature value) and blue (low feature value) encode the direction of influence on risk prediction.

The SHAP analysis reveals that historical academic performance metrics (GPA, cumulative grade measures) and behavioral indicators (absences) are the dominant drivers of risk prediction, consistent with findings from the broader EDM literature [8], [12]. Demographic features (age, sex) contribute to the prediction but with substantially lower magnitude, validating the temporal checkpoint design: models gain the most predictive power at Weeks 8 and 12 when behavioral data becomes available.

2) Local Explainability: Per-Student Analysis: Fig. 8 presents representative SHAP waterfall plots for students at each risk tier, illustrating how individual feature values contribute to each student's specific risk prediction.



(a) High-risk student: elevated absences and low GPA drive the prediction upward.

(b) Low-risk student: strong GPA and low absences are protective factors.

Fig. 8: Local SHAP waterfall plots decomposing individual risk predictions into per-feature contributions.

3) Error Analysis: False Positive Investigation: To demonstrate the diagnostic value of SHAP, we analyzed a "wrongly confident" false positive — a student whom the model incorrectly classified as high-risk with high confidence.

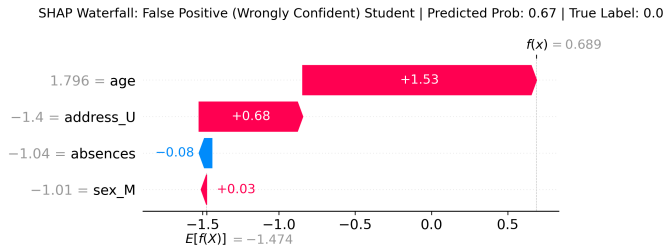


Fig. 9: SHAP waterfall plot for a false-positive case. The model over-weighted historical failure indicators while undervaluing protective factors such as high study time and extracurricular engagement.

As shown in Fig. 9, the model over-weighted historical failure indicators while simultaneously undervaluing protective factors (such as high study time). This analysis illustrates precisely why XAI is essential: without SHAP decomposition, this misclassification would have resulted in unnecessary intervention resource allocation. With SHAP, the advisor can review the feature attribution, identify the model's reasoning error, and make an informed override decision — keeping the human advisor "in the loop."

Addressing RQ1: The combination of SHAP global and local analysis confirms that academic performance metrics (GPA, grades), behavioral indicators (absences),

and historical failure records constitute the most predictive feature combination for academic probation risk. Demographic features are contributory but secondary.

H. Dashboard Visualization

An interactive Streamlit dashboard operationalizes the analytical pipeline, providing academic advisors with real-time access to: (1) a risk distribution overview (donut chart by intervention tier), (2) system F1-Macro evolution across checkpoints (line chart tracking predictive power over time), (3) global risk drivers (SHAP-based horizontal bar chart of the top 10 features), and (4) an actionable student roster (filterable table listing each student with risk probability, intervention tier, and prescribed recommended action).

V. Discussion

A. Synthesis of Research Questions

RQ1 — Optimal Feature Combination: The SHAP-driven analysis (Section IV) identifies academic performance (GPA, cumulative grades) and behavioral indicators (absences) as the dominant predictive features. Demographic variables contribute but are insufficient alone — a finding that validates the temporal checkpoint design, where behavioral features at Weeks 8 and 12 substantially amplify predictive power beyond the Week 4 demographic baseline.

RQ2 — Earliest Viable Intervention Point: The temporal checkpoint analysis demonstrates that behaviorally enriched models at Week 8 achieve F1-Macro scores in the 0.63–0.67 range (Decision Tree, Random Forest), providing approximately 8 weeks of intervention lead time before semester completion. While Week 12 models achieve the highest accuracy for Logistic Regression (0.616), the marginal accuracy gain must be weighed against the reduced intervention window. For institutions prioritizing intervention timeliness, Week 8 represents the best accuracy-intervention tradeoff observed in our experiments.

RQ3 — Warehouse vs. Traditional Approaches: ProGrade's warehouse-integrated pipeline offers structural advantages that flat-file approaches cannot replicate: (a) source-level segmentation enabling cross-dataset benchmarking, (b) surrogate key–based auditability of every prediction's data lineage, (c) conformed dimensions ensuring consistent feature definitions across institutional data sources, and (d) ETL separation of concerns isolating data quality issues from model performance analysis. While a direct quantitative comparison against a flat-file baseline was not conducted in the current study, the cross-dataset benchmark (Section IV-F) provides concrete evidence that the warehouse's source-tagging mechanism is essential for reproducible domain-shift diagnostics.

B. The Warehouse Advantage

The star-schema data warehouse is not merely an architectural convenience — it is a prerequisite for institutional deployment. Academic institutions operate under strict data governance requirements. By structuring the analytical pipeline around a formal warehouse with conformed dimensions and traceable ETL lineage, ProGrade satisfies the auditability, reproducibility, and scalability criteria that raw ML pipelines fundamentally lack.

The cross-dataset benchmark (Section IV-F) provides a concrete illustration: the warehouse's source-tagging mechanism (via the school field in DimCourse) enables seamless segmentation of training and evaluation data by institutional origin — an operation that would require ad-hoc and error-prone DataFrame manipulations in a flat-file pipeline.

C. Explainability as a Deployment Enabler

The integration of SHAP transforms ProGrade from a predictive tool into a diagnostic instrument. The false-positive analysis (Fig. 9) demonstrates that advisors equipped with per-student feature attribution can distinguish between genuinely at-risk students and model misclassifications — a capability that is essential for maintaining institutional trust in algorithmic decision-support [14], [15].

The automated generation of PDF advisor reports (via the shap_reporter module) operationalizes this explainability, eliminating the need for advisors to interpret SHAP plots directly and instead delivering actionable, human-readable recommendations aligned with the three-tier intervention protocol.

D. Limitations

Despite its contributions, ProGrade has several limitations that must be acknowledged:

1) Simulated temporal checkpoints. The mid-semester checkpoint data is generated via programmatic label assignment rather than being collected from a real-time LMS with true temporal progression. Consequently, the features available at "Week 4" are not a genuine temporal subset of those at "Week 12" — all features exist at all checkpoints, and the temporal effect is approximated via feature-masking heuristics. While the simulation validates the conceptual framework and demonstrates the accuracy–intervention tradeoff, the reported temporal dynamics should be interpreted as directional evidence rather than precise temporal measurements. Deployment to a live system would require integration with real-time LMS data feeds.
2) Feature heterogeneity across datasets. The three data sources have different feature schemas, and the unification process introduces missing values and approximate mappings — contributing to the modest overall F1-Macro scores and negative Cohen's κ for ensemble models. A production deployment targeting a single

institution would likely achieve substantially higher accuracy.

3) Binary risk label engineering. Risk labels were engineered from source-specific grade thresholds (G3 < 10, GradeClass ≥ 3, Target = "Dropout"), which may not precisely correspond to each institution's formal definition of academic probation.
4) Absence of unstructured data. The current pipeline operates exclusively on structured tabular data. Student forum posts, LMS interaction logs, email communication patterns, and other unstructured data sources — which may contain early behavioral signals — are not incorporated.
5) Intervention outcome evaluation. ProGrade generates intervention recommendations but does not measure their downstream effectiveness (i.e., whether intervened students actually improve). A randomized controlled trial would be necessary to establish causal efficacy.
6) Small test-set sizes. The multi-checkpoint stratified split yields small per-checkpoint test sets ($n \approx 30$–40), limiting the statistical power of model comparisons. Observed F1-Macro differences below 0.10 may not reach significance.

E. Future Work

Several directions for future research emerge from this study:

- Real-time LMS integration: Connecting the warehouse to live LMS data feeds (e.g., Moodle, Canvas) to enable true real-time checkpoint scoring rather than post-hoc simulation.
- NLP for unstructured data: Incorporating Natural Language Processing (NLP) techniques to analyze student forum interactions, assignment text quality, and communication sentiment as supplementary behavioral features.
- Domain adaptation: Applying transfer learning and domain adaptation techniques to mitigate the generalization gap identified in the cross-dataset benchmark, enabling models to transfer across institutions with reduced recalibration overhead [13].
- Fairness auditing: Systematically evaluating whether the model's predictions exhibit bias with respect to protected demographic attributes (gender, socioeconomic status, ethnicity) and implementing debiasing techniques where necessary.
- Intervention efficacy study: Conducting a randomized controlled trial in partnership with an institution to measure the causal impact of ProGrade-generated interventions on student retention outcomes.
- Statistical validation at scale: Replicating the checkpoint protocol on larger institutional datasets to enable rigorous 5×2 cross-validated paired t-tests between model pairs.

F. Ethical Considerations

This study exclusively utilizes publicly available, anonymized benchmark datasets (UCI Machine Learning Repository and Kaggle). No personally identifiable student information was collected, processed, or stored at any stage of the research. All datasets are distributed under open licenses that permit research use. The data warehouse was implemented locally and contains only the pre-anonymized records from these public sources. We acknowledge that deployment of such systems in institutional settings would require compliance with applicable data protection regulations (e.g., FERPA, GDPR) and institutional review board approval before processing real student records.

VI. Conclusion

This paper presented ProGrade, a comprehensive Early Warning System for academic probation prediction that proposes a novel architectural framework for Educational Data Mining through five interrelated contributions. By integrating a star-schema data warehouse with ensemble machine learning, temporal checkpoint scoring, cross-dataset benchmarking, three-tier risk stratification, and SHAP-driven explainability, ProGrade transforms the role of educational data from a passive historical record into an active decision-support instrument.

Our experimental evaluation demonstrates that mid-semester behavioral features significantly amplify predictive power beyond demographic baselines, with the Week 8 checkpoint offering the best observed tradeoff between accuracy and intervention lead time. The cross-dataset benchmark exposes a critical generalization gap that underscores the dangers of deploying pre-trained models across institutional boundaries without recalibration. The SHAP explainability layer bridges the trust gap between algorithmic prediction and human decision-making, enabling academic advisors to make informed, evidence-based intervention decisions.

ProGrade establishes a replicable architectural blueprint for institutions seeking to deploy data-driven student retention systems that are not only accurate but auditable, explainable, and actionable.

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